BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 219 · Dijkstra: shortest paths II

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What vertex does Dijkstra settle next?

Q2. What does a weighted edge record?

Q3. What does a predecessor in a path mean?

Q4. Distance to A=3, edge A–B=4, current distance B=10. Update B.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

predecessor table — For each vertex, stores the prior vertex on its best-known path.

relaxation — Replace a distance when a cheaper route is discovered.

settled — Shortest distance is final for a vertex under non-negative edge weights.

infinity — Marker for an as-yet unreachable vertex.

route reconstruction — Follow predecessor links backwards then reverse.

New learning

Track improvements — Relax each edge by comparing the candidate total with the current best-known distance.

Remember predecessors — When a distance improves, store the vertex from which the improvement came.

Finish with the best distance — Here E reaches distance 7 after A, C, B and D have settled.

Reconstruct backwards — Follow E←D←B←C←A, then reverse it to get the forward route.

Handle unreachable vertices — A vertex remaining at infinity cannot be reached from this start along the graph's edges.

Worked example

Find the shortest route S→T and reconstruct it.

1. Predict the next state — Start S=0: tentative A=2, B=6.

2. Update the table — Settle A=2; B improves to 4 and T receives 9.

3. Update the table — Settle B=4; T improves to 5.

4. Update the table — Trace T←B←A←S, then reverse: S→A→B→T at cost 5.

We Do · Practise with guidance

Find the shortest route P→T and reconstruct the predecessors.

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Edges: P-Q=4, P-R=1, R-Q=2, Q-T=3, R-T=10.
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Predict the next state

Start P=0 gives Q=4, R=1; settle R next.

Update the table

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

R improves Q to 3 and gives T=11.

Q improves T to 6.

T←Q←R←P reverses to P→R→Q→T, cost 6.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Starting at U, edges U–V=4, U–W=1 and W–V=2. Find the shortest U→V path and total cost.

Question 2

Edges S–A=5, S–B=1, B–A=2, A–T=3 and B–T=9. Find S→T shortest distance, route and predecessor of T.

Question 3

A graph contains a disconnected vertex Z. Explain what happens to tentative distance and predecessor for Z in Dijkstra, and when the algorithm may stop.

Exit Ticket · Answer before checking

A predecessor table should be followed from the destination backwards before reversing.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. The unsettled vertex with the smallest current tentative distance.

Q2. The cost of travelling between two connected vertices.

Q3. The previous vertex on that route.

Q4. Candidate 3+4=7, so update B to 7 and predecessor(B)=A.

You Do pages · Independent questions

Q1. [4] U→W costs 1 [1]; W→V adds 2 [1]; total 3 [1]; it is shorter than the direct cost 4 [1].

Q2. [6] Start B=1, A=5 [1]; settle B=1 [1], improve A=3 [1]; settle A=3, T improves to 6 [1]; route S→B→A→T [1]; predecessor(T)=A [1].

Q3. [6] Z is unreachable [1], remains at ∞ [1] and has no valid predecessor [1]; once all unsettled vertices have ∞ distance [1], no reachable vertex remains [1] and processing can terminate [1].

Exit Ticket · Model answer

True. The predecessor of each node gives the step immediately before it on the path.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first.

GREEN PEN Then: tick correct answers; rewrite mistakes.